Semantic Understanding and Command Following

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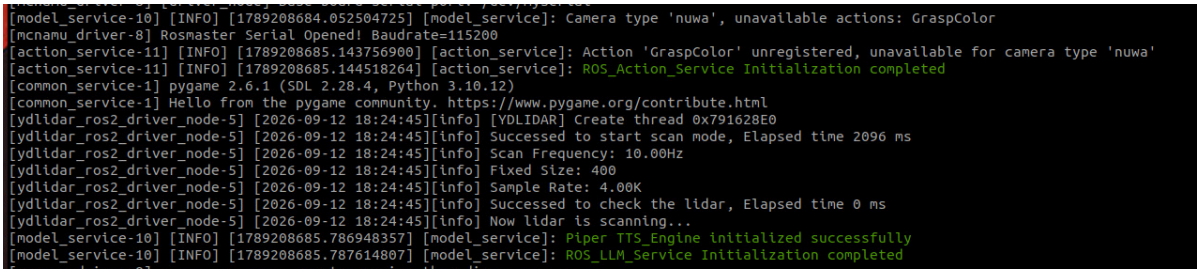
1. Course Content
2. Preparation
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5. Common Problem Troubleshooting
    5.1 Adjusting the Microphone Recording Sensitivity

1. Course Content

1. After running the large model program, the user can interact with the robot through voice conversation. The user's voice command is first converted into text through the speech recognition large model, and then the text generation large model and visual multimodality accurately understand the user's command and voice. Finally, the robot completes the specified action according to the user's command and replies to the user.

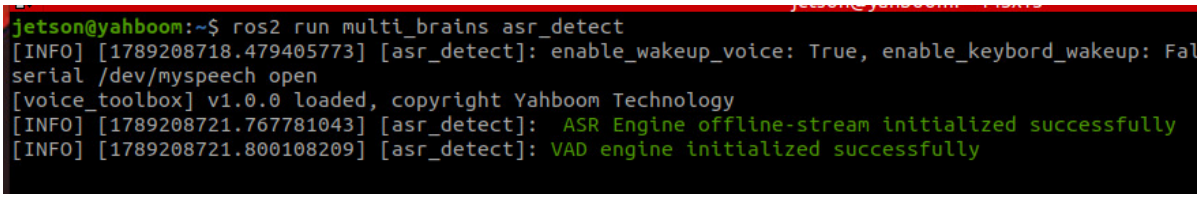
2. Preparation

```
ros2 launch yahboomcar_bringup dify_llm_agents.launch.py use_tts:=true
```



- Voice recognition node

```
ros2 run multi_brains asr_detect
```



3. Case Demonstration

A reference test case is given below. You can write your own test commands as you like.

- Please move forward 1 meter, then backward 0.5 meters, then turn left 30 degrees, turn right 90 degrees, then translate left 0.5 meters, and then translate right 10 centimeters.

```
[model_service-10] [INFO] [1792115518.675585716] [model_service]: Decision-making level planning: 1. Move forward 1 meter
[model_service-10] 2. Move backward 0.5 meters
[model_service-10] 3. Turn left 30 degrees
[model_service-10] 4. Turn right 90 degrees
[model_service-10] 5. Translate left 0.5 meters
[model_service-10] 6. Translate right 0.1 meters
[model_service-10] [INFO] [1792115518.680003279] [model_service]: "action": ['move("forward",1)'],"response": Got it, moving forward one meter now.
```

4. Code Analysis

This lesson applies the program framework of the basic AI embodied intelligence gameplay. For the code analysis, refer to the section "AI Model Basics - Core Source Code Analysis of the Architecture".

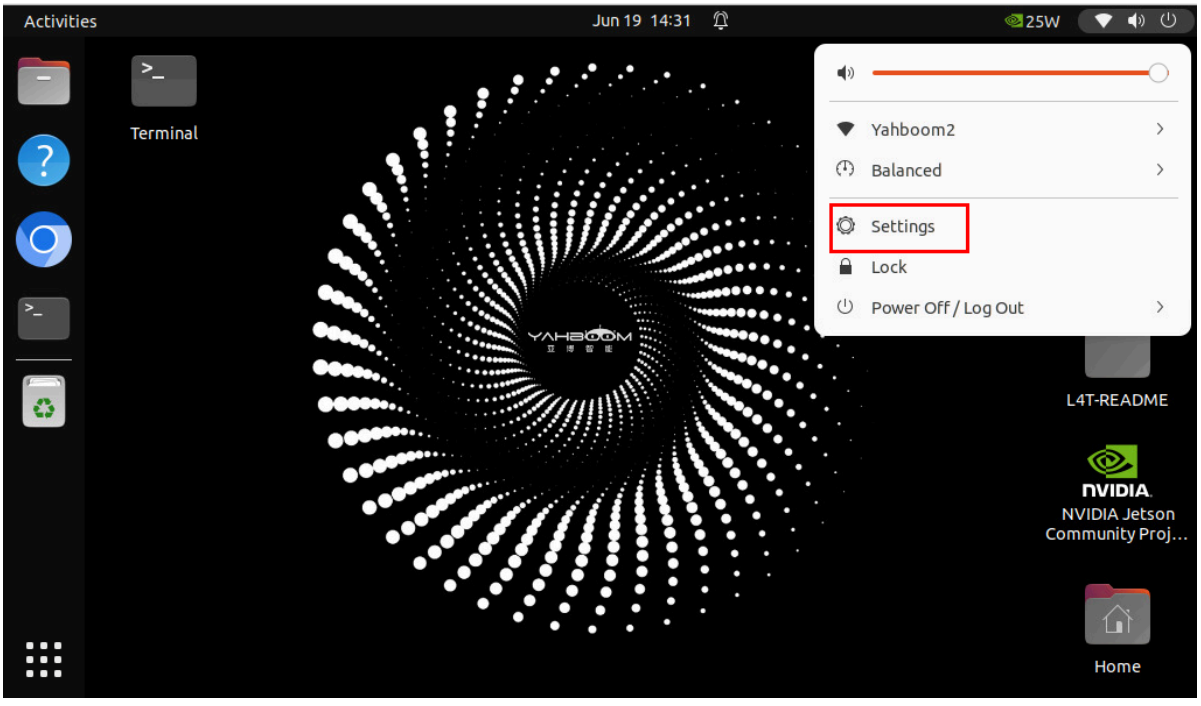
5. Common Problem Troubleshooting

5.1 Adjusting the Microphone Recording Sensitivity

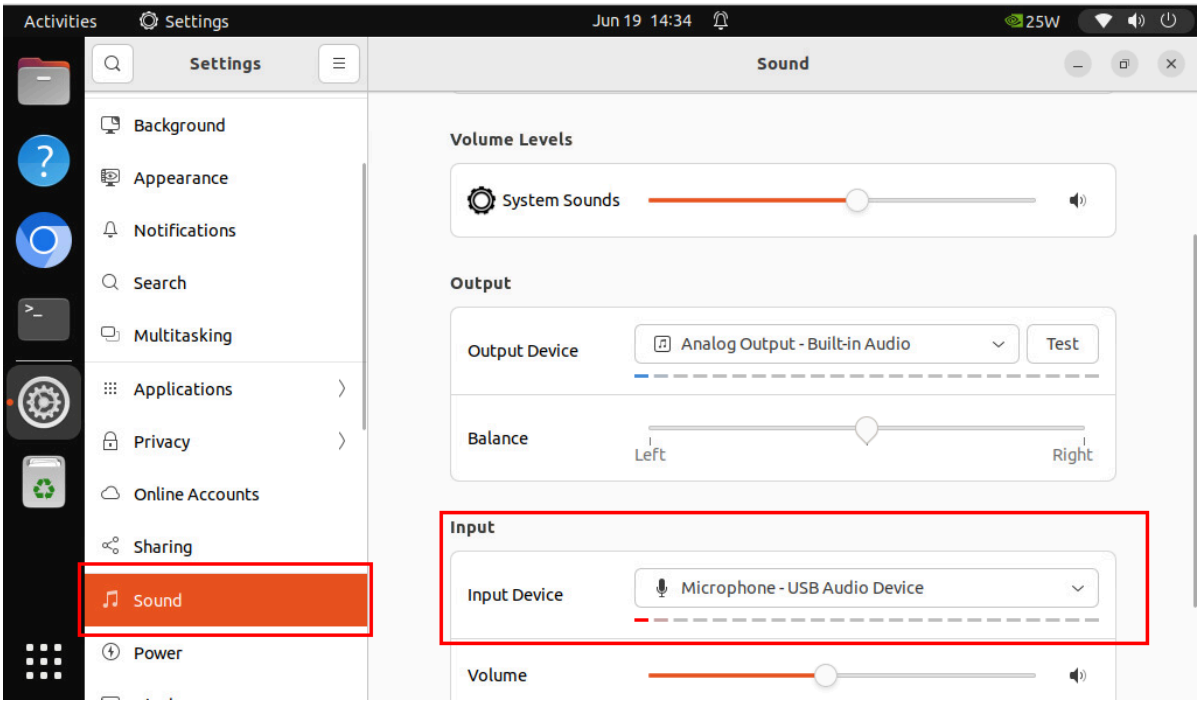
- Give priority to adjusting the microphone sensitivity; in general, the sensitivity is rarely adjusted at the software level.

First, connect to the robot's host computer screen through VNC, click the option bar in the upper right corner, and find the **Settings** option.

> **Note**
>
> - The operation method is similar for the Raspberry Pi and RDK models: both adjust the microphone sensitivity in the settings in the upper right corner, with only a slight difference in UI style.



Pull down the **Settings** list on the left, find the **Sound** option, and on the **Sound** page find **Input Device** under **Input**, then drag **Volume** below to adjust the sensitivity. Try to adjust it to a suitable value while recording.


Activities
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Balance
Left
Right
Input
Input Device
Microphone - USB Audio Device
Volume